

Solutions to JEE Main – 16 | JEE - 2022

PHYSICS

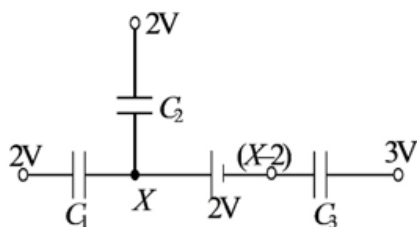
SECTION-1

1.(D) $2(X-2) + 2(X-2) + 2(X-2-3) = 0$

$$6X = 18$$

$$X = 3V$$

Hence charge on any capacitor is not zero.



2.(C) $5(0.5 + 0.3 + 0.2) = (I - 5)(0.1)$

$$I - 5 = 50$$

$$I = 55 \text{ mA}$$

3.(B) Potential energy of the system when charge Q is at O is

$$U_0 = \frac{qQ}{a} + \frac{qQ}{a} = \frac{2qQ}{a}$$

When charge Q is shifted to position O' , the potential energy will be

$$U = \frac{qQ}{(a+x)} + \frac{qQ}{(a-x)} = \frac{qQ(2a)}{(a^2 - x^2)} = \frac{2qQ}{a} \times \left(1 - \frac{x^2}{a^2}\right)^{-1}$$

$$= \frac{2qQ}{a} \times \left(1 + \frac{x^2}{a^2}\right) \quad (\because x \ll a)$$

$$\therefore \Delta U = U - U_0 = \frac{2qQ}{a} \left(1 + \frac{x^2}{a^2}\right) - \frac{2qQ}{a} = \frac{2qQ}{a^3} (x^2)$$

$$\text{Hence } \Delta U = \Delta U \propto x^2$$

4.(B) If $\vec{V} = V_x \hat{i} + V_y \hat{j} + V_z \hat{k}$, $\vec{E} = E \hat{i}$, $\vec{B} = B \hat{j}$

$$\therefore \vec{F} = q \left[E \hat{i} + (V_x \hat{i} + V_y \hat{j} + V_z \hat{k}) \times B \hat{j} \right]$$

$$= qE \hat{i} + qV_x B \hat{k} + 0 - qV_z B \hat{i} = (E - V_z B) \hat{i} + V_x B \hat{k}$$

For no deviation, net force should either be zero or in the direction of velocity of particle.

$\therefore$ For $F = 0, V_z = \frac{E}{B}, V_x = 0, V_y \rightarrow$ has any value

Thus correct option is B

5.(B) When K_1 and K_3 are closed charge on capacitor and current in inductor is given by

$$q = C\varepsilon(1 - e^{-t/RC}) = 2 \times 2 \left(1 - e^{-t/0.5 \times 2}\right) = 4 \left(1 - e^{-t}\right) = 4 \left(1 - \frac{1}{e}\right) \text{ at } t = 1 \text{ sec}$$

$$I = \frac{\varepsilon}{R} \left(1 - e^{-\frac{Rt}{L}}\right) = \frac{4}{2} \left(1 - e^{-\frac{2t}{2}}\right) = 2(1 - e^{-t}) = 2 \left(1 - \frac{1}{e}\right) \text{ at } t = 1 \text{ sec.}$$

Now when K_1, K_3 are opened and K_2 is closed then it is an $L - C$ ckt and Let q_0 be maximum charge.

From conservation of energy for $L-C$ ckt

$$\frac{q_0^2}{2C} = \left(\frac{q^2}{2C} + \frac{1}{2} L I^2\right) \Rightarrow \frac{q_0^2}{2 \times 2} = \frac{4^2 \left(1 - \frac{1}{e}\right)^2}{2 \times 2} + \frac{1}{2} \times 2 \times 2^2 \left(1 - \frac{1}{e}\right)^2$$

$$\Rightarrow q_0 = 4\sqrt{2} \left(1 - \frac{1}{e}\right)$$

6.(D)
$$I = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{240}{30} = 8A$$

Reading of voltmeter is zero.

7.(B)
$$B = 10^{-7} \times \frac{\pi \times i}{r} = 10^{-7} \times \frac{\pi \times 10}{20 \times 10^{-2}} = B = 5\pi \mu T$$

8.(C) Magnetic flux through the loop A ;

$$\phi = B\pi r^2 = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}} \pi r^2$$

Induced EMF in the loop A ;

$$\varepsilon = \frac{d\phi}{dt} = \frac{\mu_0 I R^2 \pi r^2}{2} \left(-\frac{3}{2(R^2 + x^2)^{5/2}} 2x \frac{dx}{dt} \right) = -\frac{3\mu_0 I R^2 \pi r^2 v}{2} \left[\frac{x}{(R^2 + x^2)^{5/2}} \right]$$

Induced EMF is maximum when $\frac{d\varepsilon}{dx} = 0$

$$\Rightarrow (R^2 + x^2) - 5x^2 = 0 \text{ or } x = \frac{R}{2}$$

9.(B) Since voltage is lagging behind the current, presence of an inductor in the box is not a must.

10.(D) The electric field inside a conductor is zero. So if we take a Gaussian spherical surface just covering the inner surface of the shell C, the flux on it will be zero.

Hence from Gauss's law, the charge on the inner surface of shell C should be $-3Q$, and from conservation of charge, charge on the outer surface of C will be $+2Q$.

11.(B) Area vector $\vec{A} = l^2 \hat{i} \Rightarrow \phi = \vec{E} \cdot \vec{A} = al^2$

12.(A) Potential of charged sphere is given by the formula

$$V = \frac{kQ}{R}$$

Hence, by $Q = CV$,

$$C = \frac{Q}{V} = \frac{R}{k}$$

As per given information, $\frac{R}{k} = 1 \mu F$

After splitting,

$$8 \left(\frac{4}{3} \pi r^3 \right) = \frac{4}{3} \pi R^3$$

$$r = \frac{R}{2}$$

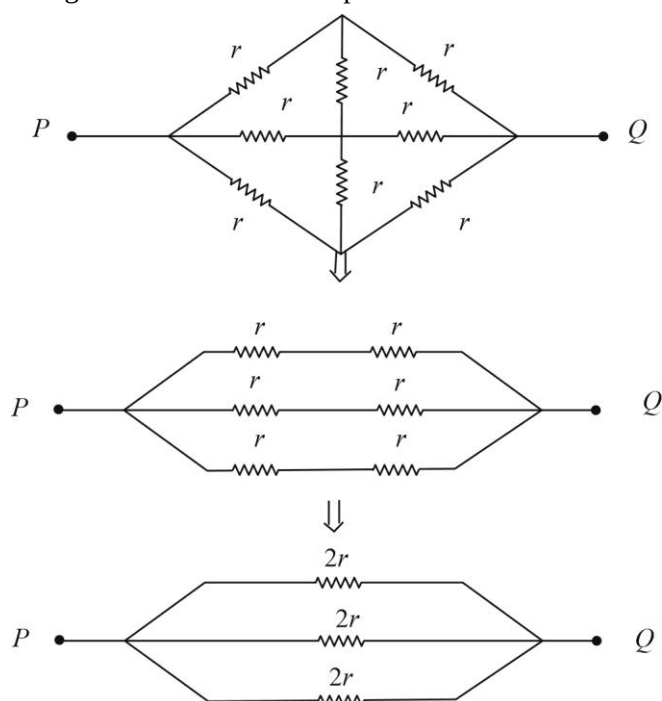
Thus, capacitance of each small drop is $V = \frac{r}{k} = \frac{1}{2} \frac{R}{k} = \frac{1}{2} \mu F$

13.(C) Magnetic moment $M = \frac{qL}{2m} = \frac{q}{2m} \frac{2}{5} m R^2 \omega \Rightarrow M = \frac{1}{5} q R^2 \omega$

14.(D) $I_1 = \frac{E}{r_1 + r}, Q_1 = I_1^2 r_1 t = \left(\frac{E}{r_1 + r} \right)^2 \times r_1 t$

$$Q_2 = \left(\frac{E}{r_2 + r} \right)^2 \times r_2 t, r = \sqrt{r_1 r_2}$$

15.(B) The given circuit can be simplified as follows



$$R' = \frac{2r}{3} = \frac{2}{3} \times \frac{3}{2} = 1\Omega \quad [\because r = \frac{3}{2}\Omega]$$

16.(B) When crosses AB

$$qvB_0 \cos \theta = mg$$

$$qvB_0 \sin \theta = qE$$

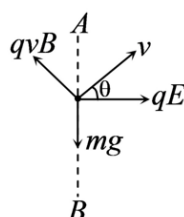
$$\tan \theta = \frac{qE}{mg} = 1$$

$$\theta = \frac{\pi}{4}$$

Along horizontal $v \cos \theta = \frac{qE}{m} t_0$

$$u - gt_0 = v \sin \theta$$

$$u = \left(g + \frac{qE}{m} \right) t_0 = 2gt_0$$



17.(D) $\phi = 6t^2 + 7t + 1 \Rightarrow \frac{d\phi}{dt} = 12t + 7$

At time, $t = 2s$

$$\frac{d\phi}{dt} = 24 + 7 = 31 \text{ volt}$$

Direction of current is from left to right according to Lenz's law and the Flemming's right hand rule.

18.(B) Electron doesn't experience a force in x-direction thus momentum in x-direction is conserved.

Therefore,

$$\Rightarrow v \cos \beta = u \cos \alpha$$

$$\therefore \frac{K_{in}}{K_{out}} = \frac{\frac{1}{2}mu^2}{\frac{1}{2}mv^2} = \frac{\cos^2 \beta}{\cos^2 \alpha}$$

19.(A) The growth of current is given by $i = i_0(1 - e^{-t/\tau}) \Rightarrow \frac{di}{dt} = \frac{i_0}{\tau} e^{-t/\tau}$

Energy stored in the form of magnetic field energy is: $U_B = \frac{1}{2}Li^2$

Rate of increase of magnetic field energy is: $R = \frac{dU_B}{dt} = Li \frac{di}{dt}$

This will be maximum when $\frac{dR}{dt} = 0 \Rightarrow e^{-t/\tau} = 1/2$

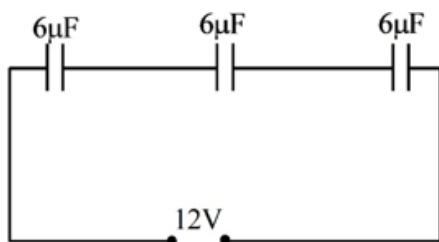
Substituting: $R_{\max} = \frac{Li_0^2}{\tau}$

20.(A) $R = \frac{mv}{qB}$, $2R < r \frac{1}{2} \Rightarrow V < \frac{qBR}{M} \Rightarrow V < \frac{qBr}{2M} = \frac{qr}{2M} \mu_0 ni$

SECTION-2

21.(4) $1\mu F$, $2\mu F$ and $3\mu F$ are in parallel and their equivalent capacitance is $6\mu F$.

Hence, each $6\mu F$ capacitor has potential difference of 4V.



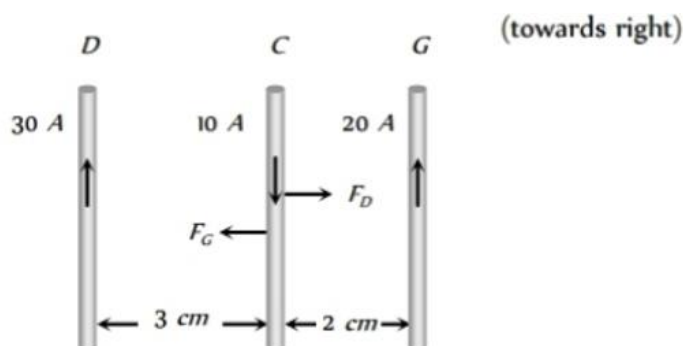
22.(22) Current required by each bulb, $I = \frac{P}{V} = \frac{100}{220}$ amp

If n bulbs are joined in parallel,

Then $nI = I_{fuse}$ or $n \times \frac{100}{220} = 10$ or $n = 22$

23.(0) Force on wire C due to wire D

$$F_D = 10^{-7} \times \frac{2 \times 30 \times 10}{2 \times 10^{-2}} \times 25 \times 10^{-2} = 5 \times 10^{-4} N$$



Force on wire C due to wire G

$$F_D = 10^{-7} \times \frac{2 \times 20 \times 10}{2 \times 10^{-2}} \times 25 \times 10^{-2} = 5 \times 10^{-4} N \quad (\text{Towards left})$$

$\Rightarrow$ Net force on wire C is $F_{net} = F_D - F_G = 0$

24.(10) $V_{rms} = \sqrt{\frac{1}{T} \int_0^T 10^2 dt} = 10V$

25.(3) Net Electric field $\vec{P}$

$\vec{E} = \vec{E}_1 + \vec{E}_2$ due to vertical wire

$\vec{E}_1 = \frac{\lambda_1}{2\pi\epsilon_0 y}$ in $+y$ direction, $\vec{E}_2 = \frac{\lambda_2}{2\pi\epsilon_0 x}$ due to vertical wire in x direction.

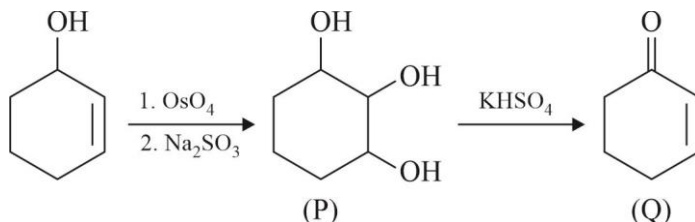
$$\alpha \text{ angle of electric field with } x \text{ direction } \tan \alpha = \frac{E_1}{E_2} = \frac{\lambda_1 / 2\pi\epsilon_0 y}{\lambda_2 / 2\pi\epsilon_0 x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{\lambda_1 x}{\lambda_2 y} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{x}{y} \sqrt{3};$$

$$\frac{\lambda_1}{\lambda_2} = 3$$

CHEMISTRY**SECTION-1**

1.(D) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $[\text{Cu}(\text{H}_2\text{O})_4]^{2+} \text{SO}_4^{2-} \cdot \text{H}_2\text{O}$ contains coordinated and H-bonded water and its unit cell is triclinic.

2.(C)



(P) on reaction with KHSO_4 undergoes pinacol-pinacolone rearrangement followed by dehydration.

3.(C) Methionine (sulphur containing) is naturally occurring α – amino acid and it is an essential amino acid. The configuration of naturally occurring amino acid is L-configuration of asymmetric alpha carbon.

4.(D) Dissolution of acid and its ionization in water is exothermic.

5.(A) (A) $\text{NaCl} + \text{MnO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{MnSO}_4 + \text{Na}_2\text{SO}_4 + \text{Cl}_2 + \text{H}_2\text{O}$

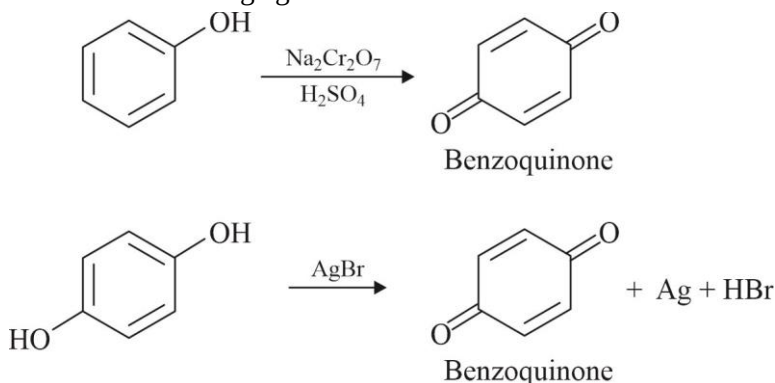
(B) $\text{NH}_3 + \text{CaOCl}_2 \longrightarrow \text{CaCl}_2 + \text{N}_2 + \text{H}_2\text{O}$

(C) $\text{NaCl} + \text{H}_2\text{SO}_4 \longrightarrow \text{NaHSO}_4 + \text{HCl}$

(D) $\text{NaCl} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{K}_2\text{SO}_4 + \text{CrO}_2\text{Cl}_2 + \text{H}_2\text{O}$

6.(B) It acts as p-type semiconductor due to the presence of hole because of formation of Cu^{2+} ions.

7.(A) Phenols are reducing agent.

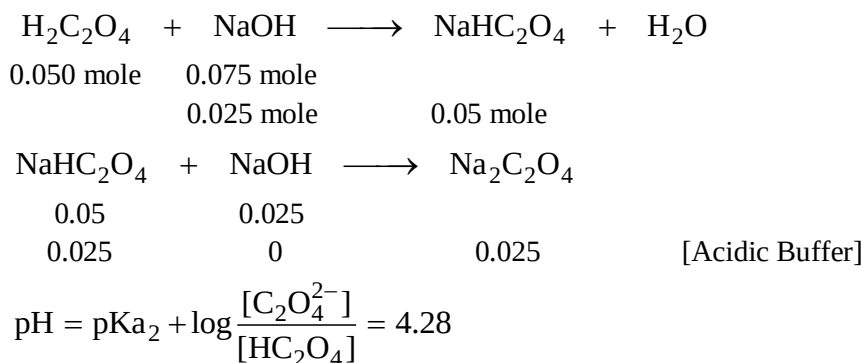


8.(C) The main force which stabilize the 2° and 3° structures of proteins are hydrogen bonds, disulphide linkages, Vander Waal's and electrostatic forces of attraction.

9.(B) $\Delta T_f = i \times k_f \times m$

$$1.30 = 1.4 \times 1.86 \times m$$

$$M = m = \frac{1.30}{1.40} \times \frac{1}{1.86} = 0.499 \approx 0.5$$

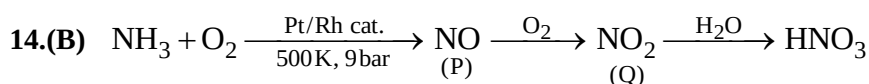


- 10.(D) (A) $\text{Br}_2 + \text{Na}_2\text{CO}_3 \longrightarrow \text{NaBr} + \text{NaBrO}_3 + \text{CO}_2$
 (B) $\text{XeF}_6 + \text{NaOH} \longrightarrow \text{Na}_4\text{XeO}_6 + \text{Xe} + \text{NaF} + \text{H}_2\text{O}$
 (C) $(\text{CN})_2 + \text{KOH} \longrightarrow \text{KCN} + \text{KCNO} + \text{H}_2\text{O}$
 (D) $\text{Br}_2 + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Na}_2\text{SO}_4 + \text{NaBr}$

11.(C) $2.5 = \frac{1 \times M_0}{(415 \times 10^{-10})^3 \times 6 \times 10^{23}}$
 $M_0 = 107.21 = 40 + 10.91x \Rightarrow x = 6.16 \approx 6$

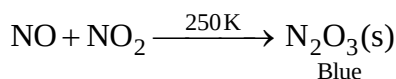
- 12.(C) Glucose undergo fermentation in the presence of enzyme, zymase, which is found in yeast. Fermentation takes place in anaerobic conditions i.e. in absence of air and it result in formation of $\text{C}_2\text{H}_5\text{OH}$ and CO_2 , if air gets into fermentation mixture, the oxygen of air oxidises ethanol to ethanoic acid which in turn destroys the taste of alcoholic drinks?

- 13.(C) Vitamine $\text{B}_{12} \Rightarrow$ RBC deficient in hemoglobin
 Vitamine E $\Rightarrow$ Increased fragility of RBCs
 Vitamine K $\Rightarrow$ Increased blood clotting time



HNO_3 and water do not form ideal solution.

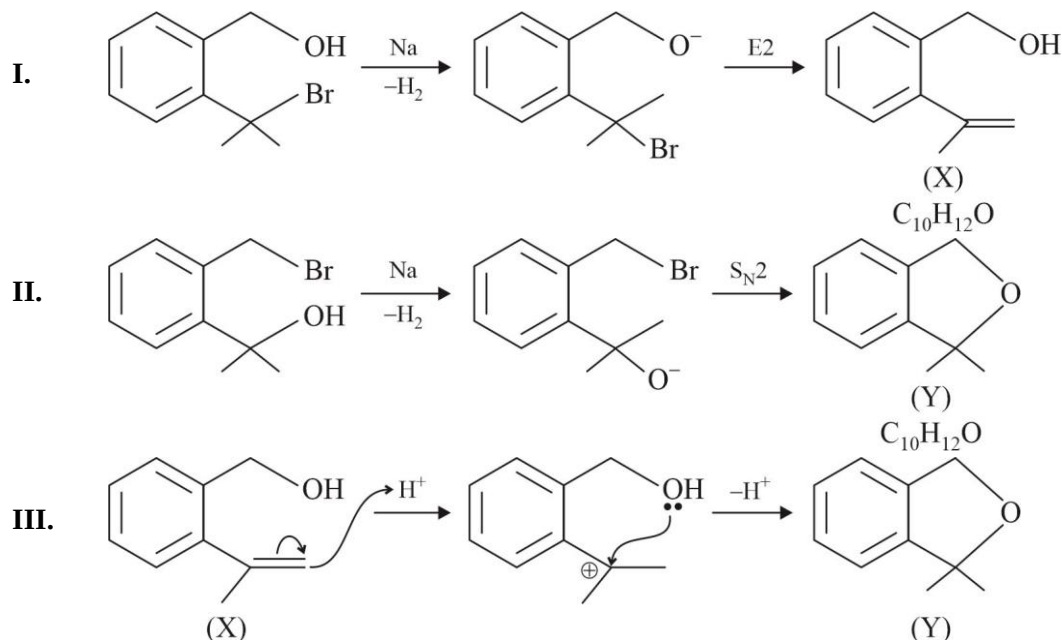
This mixture shows negative deviation from ideal behaviour and it form azeotropic mixture on distillation. NO and NO_2 are paramagnetic.



- 15.(D) F $\Rightarrow$ Only one non-zero oxidation state
 I $\Rightarrow$ Both positive and negative oxidation state
 Ne $\Rightarrow$ Only zero oxidation state
 Xe $\Rightarrow$ Only positive oxidation state

- 16.(C) LiH $\Rightarrow$ Ionic solid
 CH_4 $\Rightarrow$ Non polar molecular solid
 H_2O $\Rightarrow$ H-bonded molecular solid
 Xe $\Rightarrow$ Non-polar molecular solid

17.(C)

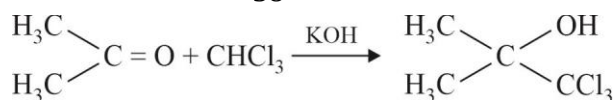


18.(A) It is a nucleotide. Nucleotides, the monomer units of RNA and DNA, consist of a pentose sugar (ribose in case of RNA) and (2-deoxyribose in case of DNA) a phosphate group and a nitrogenous base.

19.(D) Acetone and chloroform are miscible and do not form ideal liquid solution. This mixture undergoes negative deviation from ideal behaviour.

$$\text{Mole of } \text{CHCl}_3 = \frac{1500}{120} = 12.5 \quad (\text{Limiting reagent})$$

$$\text{Mole of acetone} = \frac{800}{58} = 13.79$$



$$\text{Mass of product formed} = 12.5 \times 178 = 2225 \text{ gm}$$

20.(C) S–S(monoclinic sulphur) → Single bond = 204 pm

S–S(Cyclo-S₆-chair form) → Single bond = 205.7 pm

S–S(diatomic sulphur) → Double bond

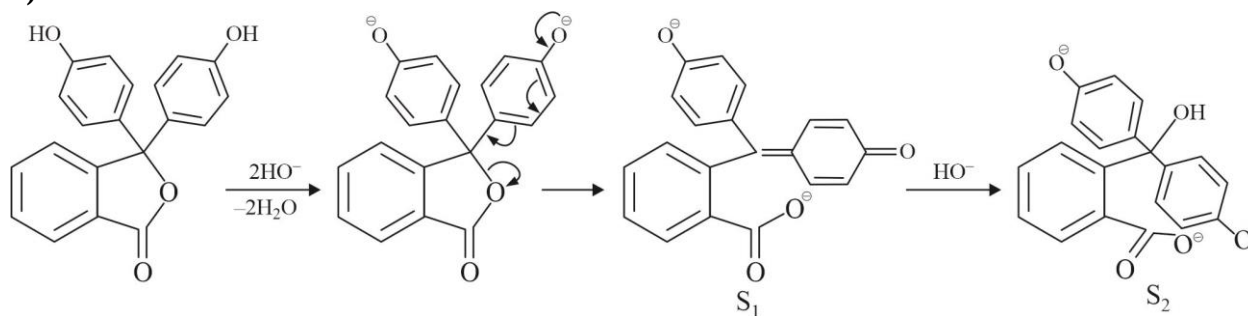
SECTION-2

21.(20) Coordination number of Mn = 6

Coordination number of La = 12

$$\text{Coordination number of O} = \frac{2}{20}$$

22.(27)



23.(51) Polypeptides with fewer amino acids are likely to be called proteins if they ordinarily have a well defined conformation of protein such as insulin which contains 51 amino acids.

24.(2.05)

$$\frac{P^{\circ} - P_s}{P_s} \times \frac{1000}{M_{\text{solvent}}} = m$$

$$\text{Loss in mass of pure solvent} \propto P^{\circ} - P_s$$

$$\text{Loss in mass of solution} \propto P_s$$

$$\frac{0.0685}{w} = \frac{0.725 \times 46}{1000}$$

$$\text{Loss in mass of solution, } w = \frac{0.0685 \times 1000}{0.725 \times 46} = 2.0539 \approx 2.05 \text{ gm}$$

25.(5) All statements are correct.

MATHEMATICS
SECTION-1

1.(A) $f \circ f(x) = (a - (f(x))^n)^{1/n} = (a - a + x^n)^{1/n} = |x| = x$

Clearly $g(x) - x > 0$ for all $x \in R$

$$\Rightarrow g \circ f(x) - f(x) > 0 \quad \forall x \in R$$

Adding both we get $g \circ g(x) - x > 0$

$$\Rightarrow g \circ f(x) - f \circ f(x) > 0 \quad \forall x \in R$$

so $g \circ f(x) - f \circ f(x) = 0$ has no solution.

2.(A) $f\left(x + \frac{1}{3}\right) = \left\{x + \frac{1}{3}\right\} + \left\{x + 1 - \frac{1}{3}\right\} + \{x\}$

$$= \left\{x + \frac{1}{3}\right\} + \left\{x - \frac{1}{3}\right\} + \{x\} = f(x)$$

So period $= \frac{1}{3}$

3.(B) $t = x^2 + 4k + \alpha^2 - \alpha$ for onto function $0 \leq t < \infty$

$$\Rightarrow D = 0 \Rightarrow 16 = 4(\alpha^2 - \alpha) \Rightarrow \alpha = \frac{1 \pm \sqrt{17}}{2}$$

4.(C) Clearly $\log_e \sin^{-1}[x]$ is defined only for $[x] = 1$

So range of $f(x) = \sin^{-1} \log_e 1 + \log_e \sin^{-1}(1)$

$$= \sin^{-1} 0 + \log_e (\pi/2) = \left\{ \log_e \frac{\pi}{2} \right\}$$

5.(A) $\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - x^2} \left(1 - 2 \sin^2 \frac{x}{2}\right) \left(1 - 2 \sin^2 \frac{3x}{2}\right)}{x^2}$

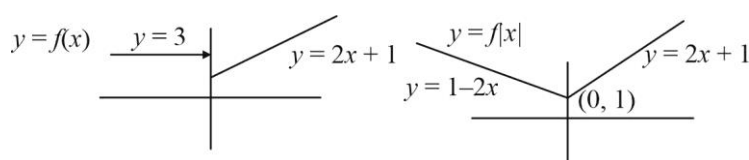
$$\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - x^2} - \frac{4\sqrt{1 - x^2} \sin^2 \frac{x}{2} \cdot \sin^2 \frac{3x}{2}}{4 \left(\frac{x}{2}\right)^2} + \frac{2 \sin^2 \frac{x}{2} \sqrt{1 - x^2}}{4 \left(\frac{x}{2}\right)^2} + \frac{2 \sin^2 \frac{3x}{2} \cdot \sqrt{1 - x^2}}{\left(\frac{3x}{2}\right)^2 \frac{4}{9}}}{x^2}$$

$$= \frac{1}{2} - 0 + \frac{1}{2} + \frac{9}{2} = \frac{11}{2}$$

$$\begin{aligned}
 6.(B) \quad \lim_{x \rightarrow 0} \frac{-\log_e \tan\left(\frac{\pi}{4} - k_1 x\right)}{\tan k_2 x} &= \lim_{x \rightarrow 0} \frac{\log_e \left(\frac{1 - \tan k_1 x}{1 + \tan k_1 x}\right)}{\tan k_2 x} \\
 &= \lim_{x \rightarrow 0} \frac{-\log_e \left(1 - \frac{2 \tan k_1 x}{1 + \tan k_1 x}\right)}{\left(\frac{-2 \tan k_1 x}{1 + \tan k_1 x}\right)} \cdot \frac{(-2 \tan k_1 x)}{(1 + \tan k_1 x)(k_1 x)} \cdot \frac{k_2 x}{\tan k_2 x} \cdot \frac{(k_1 x)}{(k_2 x)} \\
 \Rightarrow 1 &= -(-1)(-2)(1) \left(\frac{k_1}{k_2}\right) \Rightarrow k_2 = 2k_1
 \end{aligned}$$

$$\begin{aligned}
 7.(C) \quad b &= \lim_{x \rightarrow 0} \frac{f(1 - \cos x)}{(1 - \cos x)^2} \cdot \left(\frac{1 - \cos x}{x^2}\right)^2 \cdot \left(\frac{x}{\sin x}\right)^2 \cdot \frac{x^2}{g(x)} = a \cdot \left(\frac{1}{2}\right)^2 (1)^2 \cdot l \\
 \Rightarrow l &= \lim_{x \rightarrow 0} \frac{x^2}{g(x)} = \frac{4b}{a} \\
 \Rightarrow l &= \lim_{x \rightarrow 0} \frac{g(1 - \cos 2x)}{(1 - \cos 2x)^2} \cdot \left(\frac{1 - \cos 2x}{4x^2}\right)^2 \cdot \frac{16x^4}{x^4} = \left(\frac{a}{4b}\right) \left(\frac{1}{2}\right)^2 \cdot 16 = \frac{a}{b}
 \end{aligned}$$

8.(D)



Clearly $f(x)$ and $f|x|$ both are not differentiable at $x=0$

$$9.(D) \quad \text{Let } g(x) = f^{-1}(x), g \circ f(x) = x \Rightarrow g' \circ f(x) = \frac{1}{f'(x)}$$

$$\text{If } f(\alpha) = \pi \Rightarrow (2\alpha - \pi)^3 + 2\alpha - \cos \alpha = \pi \Rightarrow \alpha = \frac{\pi}{2}$$

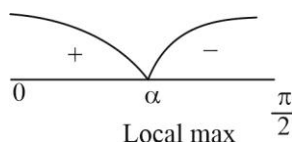
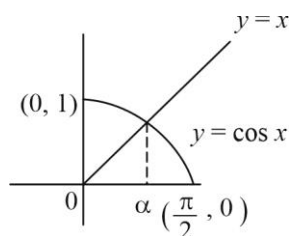
$$g' \circ f(x) = \frac{1}{f'(x)} \text{ put } x = \frac{\pi}{2}$$

$$g' \circ f\left(\frac{\pi}{2}\right) = \frac{1}{f'(\pi/2)} \Rightarrow g'(\pi) = \frac{1}{3}$$

$$10.(B) \quad f(x) = \frac{x}{1 + x \tan x} \Rightarrow f'(x) = \frac{1(1 + x \tan x) - x(\tan x + x \sec^2 x)}{(1 + x \tan x)^2}$$

$$\Rightarrow f'(x) = \frac{1 - x^2 \sec^2 x}{(1 + x \tan x)^2} = \frac{(\cos x - x)(\cos x + x)}{\cos^2 x (1 + x \tan x)^2}$$

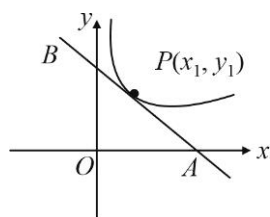
$$f'(x) = 0 \Rightarrow \cos x = x \Rightarrow x = \alpha$$



$$11.(B) \quad mx^{m-1}y + x^m \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-my_1}{x_1}$$

$$\text{Tangent at } (x_1, y_1) \quad y - y_1 = \frac{-m y_1}{x_1} (x - x_1) \text{ also } x_1^m y_1 = b^m$$

$$A \rightarrow \left(x_1 \left(\frac{m+1}{m} \right), 0 \right), \quad B \rightarrow (0, y_1(1+m))$$



$$\text{Area } \triangle AOB = \left| \frac{1}{2} x_1 y_1 \frac{(1+m)^2}{m} \right| = \left| \frac{1}{2} b^m x_1^{1-m} \frac{(1+m)^2}{m} \right|$$

Clearly area is independent of x_1 is $m=1$

$$12.(A) \quad f(x) = -\pi - 2 \tan^{-1} g(x), \quad g(x) \leq -1$$

$$= 2 \tan^{-1} g(x), \quad -1 \leq g(x) \leq 1$$

$$= \pi - 2 \tan^{-1} g(x), \quad g(x) \geq 1$$

$$f'(x) = \frac{-2g'(x)}{1+(g(x))^2}, \quad |g(x)| > 1$$

$$= \frac{2g'(x)}{1+(g(x))^2}, \quad |g(x)| < 1$$

Clearly for $|g(x)| > 1$.

If $g'(x) > 0$ then $f'(x) < 0$

$$13.(B) \quad \text{Consider } \phi(x) = f(x) - \frac{x^2}{2}, \quad \phi''(x) = f''(x) - 1$$

$$\phi(1) = \phi(2) = \phi(3) = 0$$

By Rolle's theorem $\phi'(c_1) = 0$ for at least one $c_1 \in (1, 2)$

And $\phi'(c_2) = 0$ for at least one $c_2 \in (2, 3)$

Also $\phi''(x) = 0$ for at least one $x \in (c_1, c_2)$

$$\Rightarrow f''(x) = 1 \text{ for at least one } x \in (c_1, c_2) \Rightarrow x \in (1, 3)$$

$$\Rightarrow y = \frac{3}{2} - \frac{x}{2} \text{ solve with } y^2 = x(2-x)^2$$

$$\Rightarrow \left(\frac{3}{2} - \frac{x}{2}\right)^2 = x(2-x)^2 \Rightarrow x=1, -1, \frac{9}{4}$$

Clearly $x = -1$ can not lie on curve so required point is $\left(\frac{9}{4}, \frac{3}{8}\right)$

18.(B) $(AB)^1 = A^1 B^1$

$$(AB)^2 = (AB)(AB) = A(BA)B = A(A^2B)B = A^3B^2 = A^{2^2-1}B^2$$

$$(AB)^3 = (AB)^2(AB) = (A^3B^2)(AB) = A^3B(BA)B$$

$$= A^3B(A^2B)B = A^3(BA)AB^2 = A^3(A^2B)AB^2$$

$$= A^5(BA)B^2 = A^5(A^2B)B^2 = A^7B^3 = A^{2^3-1}B^3$$

$$\text{Hence } (AB)^{10} = A^{2^{10}-1} B^{10} \Rightarrow k = 1023$$

19.(D) Let $x = \cos \theta \Rightarrow \theta = \cos^{-1} x \Rightarrow \theta \in [0, \pi]$

$$2\theta = \sin^{-1}(2 \cos \theta \sin \theta) \Rightarrow \sin^{-1} \sin 2\theta = 2\theta$$

$$\Rightarrow 2\theta \in \left[0, \frac{\pi}{2}\right] \Rightarrow \theta \in \left[0, \frac{\pi}{4}\right] \Rightarrow \cos \theta \in \left[\frac{1}{\sqrt{2}}, 1\right]$$

20.(B) Clearly $f(x)$ is into

$$f(x_1) = f(x_2) \Rightarrow \sin x_1 = \sin x_2 \Rightarrow x_2 = n\pi + (-1)^n x_1$$

$$\because x_1, x_2 \in Q \text{ so only possible } n = 0$$

$$\Rightarrow x_1 = x_2 \Rightarrow f(x) \text{ one-one.}$$

SECTION-2

21.(15) Let $x^2 + 13x + 42 = t$

$$\cot^{-1} t + \cos^{-1}(t+1) = \frac{\pi}{2} \Rightarrow t+1 \in [-1, 1] \Rightarrow t \in [-2, 0]$$

$$\Rightarrow \cot^{-1} t \geq \frac{\pi}{2} \text{ and } \cos^{-1}(t+1) \in [0, \pi]$$

$$\text{So only possible } t = 0 \Rightarrow x = -6, -7, n = 2$$

$$\Rightarrow |x_1| + |x_2| + n = 6 + 7 + 2 = 15$$

$$22.(1) \quad x \sin \alpha \cos y + x \cos \alpha \sin y = \sin y$$

$$\frac{\sin y}{\cos y} = \frac{x \sin \alpha}{1 - x \cos \alpha} \Rightarrow \sec^2 y \frac{dy}{dx} = \frac{\sin \alpha [1 \cdot (1 - x \cos \alpha) - x (-\cos \alpha)]}{(1 - x \cos \alpha)^2}$$

$$(1 + \tan^2 y) \frac{dy}{dx} = \frac{\sin \alpha}{(1 - x \cos \alpha)^2}$$

$$\left[1 + \frac{x^2 \sin^2 \alpha}{(1 - x \cos \alpha)^2} \right] \frac{dy}{dx} = \frac{\sin \alpha}{(1 - x \cos \alpha)^2} \Rightarrow \frac{dy}{dx} = \frac{\sin \alpha}{x^2 - 2x \cos \alpha + 1}$$

$$\Rightarrow m = \sin \alpha, n = -\cos \alpha \Rightarrow m^2 + n^2 = 1$$

$$23.(7) \quad A^T = ((I + B)(I - B)^{-1})^T = ((I - B)^{-1})^T (I + B)^T$$

$$= ((I - B)^T)^{-1} (I + B)^T = (I + B)^{-1} (I - B)$$

$$A^T A = (I + B)^{-1} (I - B) (I + B) (I - B)^{-1}$$

$$= (I + B)^{-1} (I + B) (I - B) (I - B)^{-1} = I I = I$$

$$\Rightarrow |A^T A| = |I| \Rightarrow |A|^2 = 1 \Rightarrow |A| = 1 \because |A| > 0$$

$$|2A| - |\text{adj } A| = 2^3 |A| - |A|^2 = 8(1) - (1)^2 = 8 - 1 = 7$$

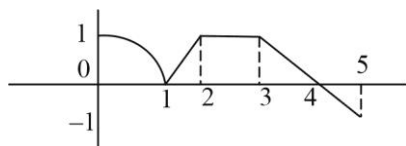
$$24.(256) \quad \Delta = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \times \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x - y)^2 (y - z)^2 (z - x)^2$$

$$\text{For } \Delta_{\min} \quad x = 2n, y = 2n + 2, z = 2n + 4$$

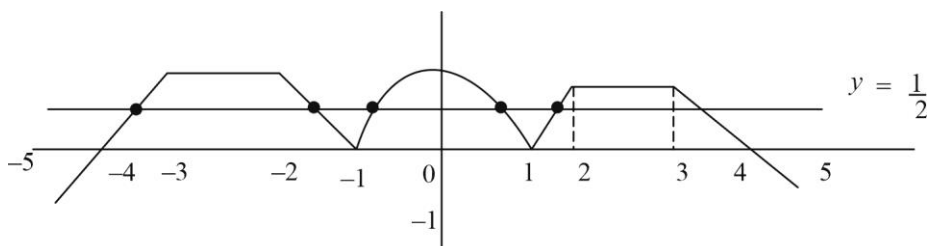
$$\Rightarrow \Delta_{\min} = (4)(4)(16) = 256$$

25.(6) Draw $y = f(1 + x)$ for $x \geq 0$. Then graph will be symmetric about y axis, $y = f(|x| + 1)$ is an even function.

For $x \geq 0$, $y = f(x + 1)$ is



So $y = f(|x| + 1)$ is



Clearly $y = \frac{1}{2}$ cut it at 6 points.